

◦ Aim:

- Experimental Verification of superposition theorem by DC excitation.

◦ Apparatus:

Equipment	Range	Quantity
Multiple Power Supply	0-30V	1
Resistors	2-2K Ω	2
Resistors	1K Ω	3
Ammeter	0-200mA	1
Breadboard	-	1
Connecting wires	-	Required

◦ Theory:

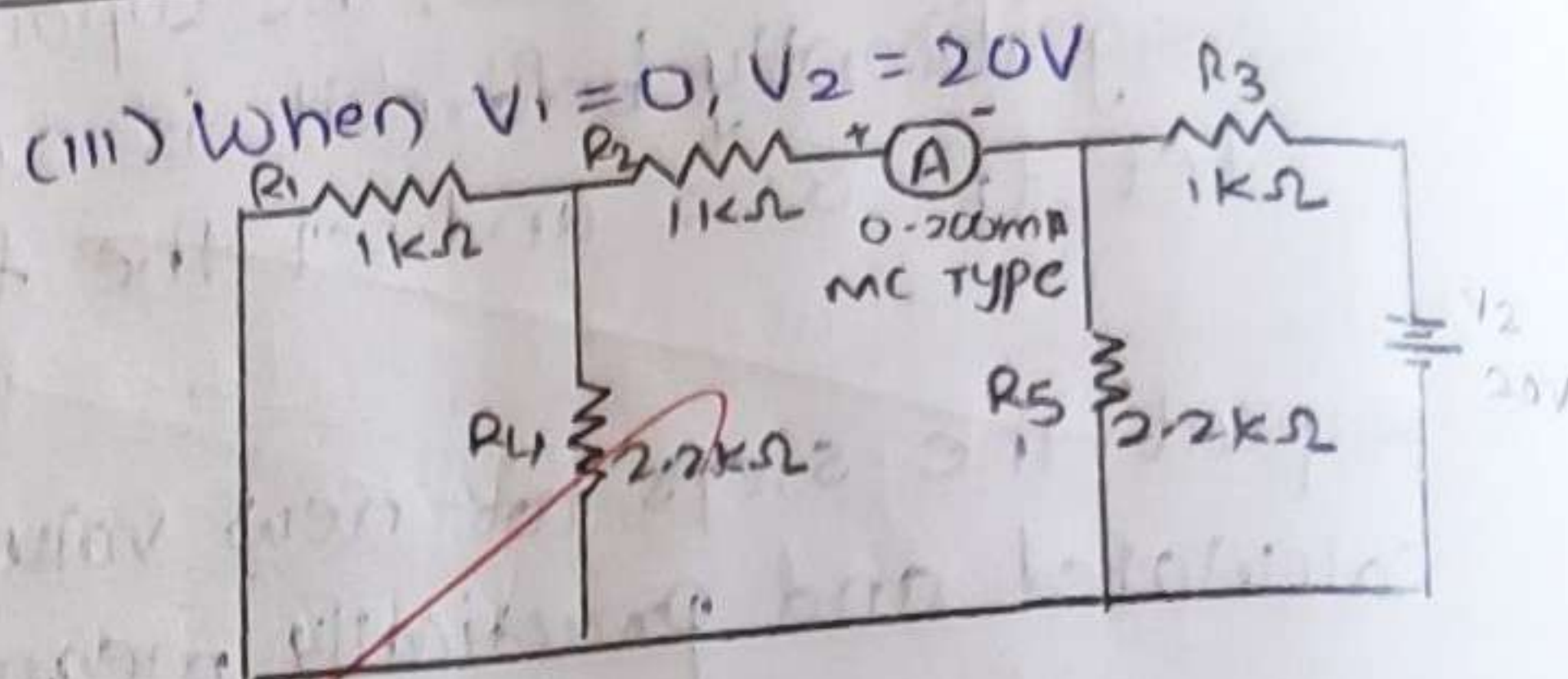
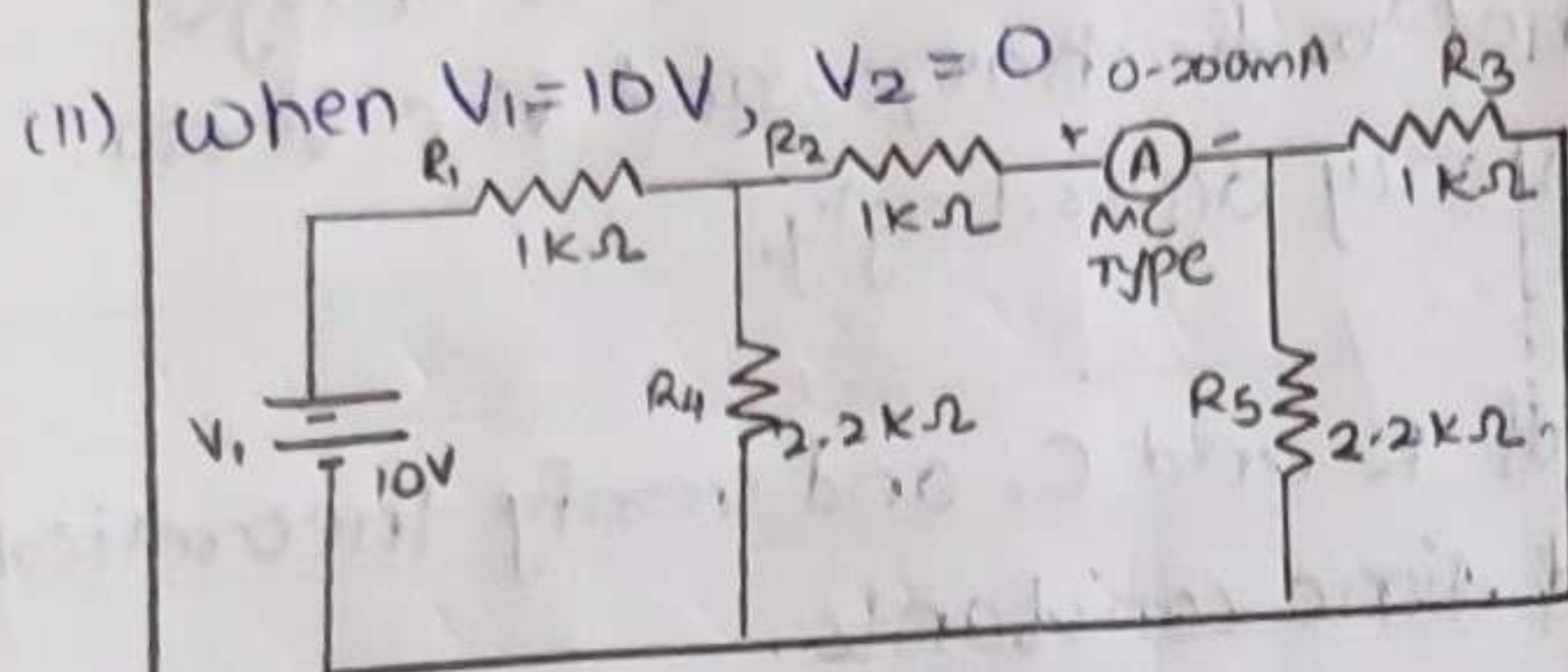
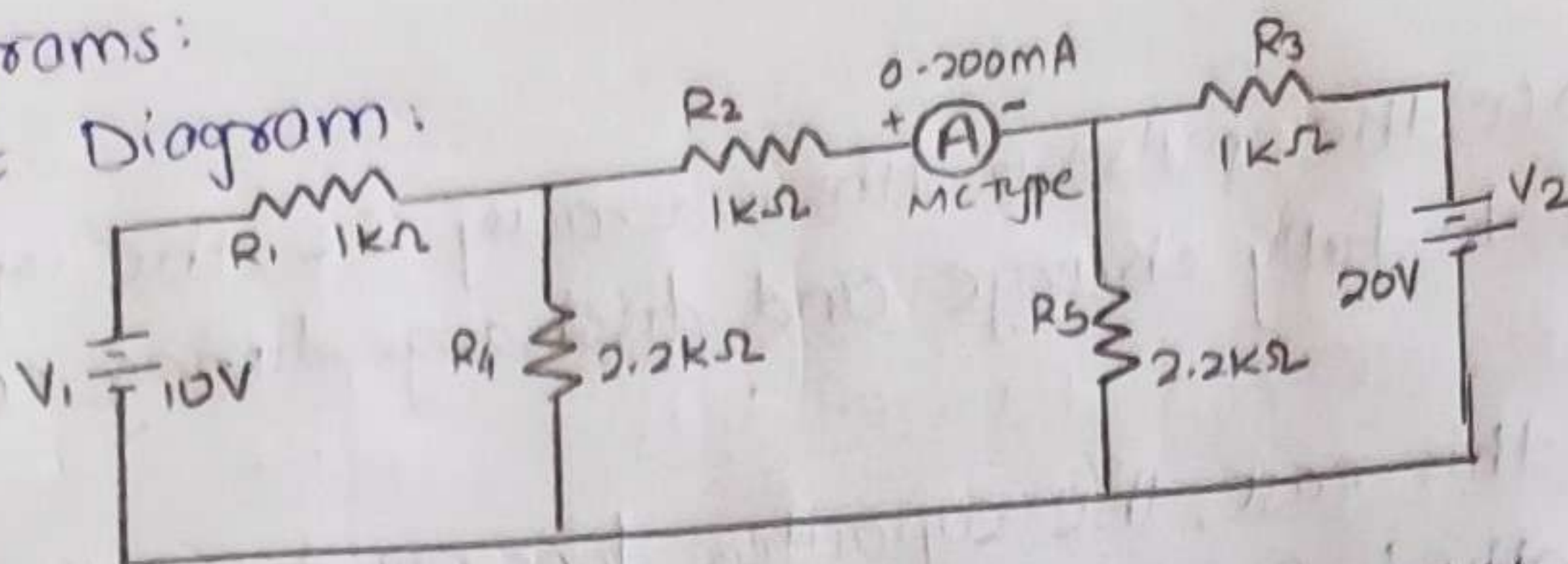
- Statement: "The superposition theorem for electrical circuits states that for a linear system, the response (voltage/current) in any branch of a bilateral linear circuit having more than one independent source equal to the algebraic sum of the responses caused by each independent source acting alone, where all the other independent sources are replaced by their internal impedances".

- To ascertain the contribution of each individual sources, all of the other sources must be 'turned off' (set to zero) by:

i) Replacing all other independent voltage sources by a short circuit (thereby eliminating potential difference i.e., $V=0$; internal impedance of ideal voltage source is zero).

• Circuit Diagrams:

(i) Main Circuit Diagram:



• Observation Table:

Case Number	Voltage Source		Current output	
	V_1	V_2	Theoretical	Experimental
1	10V	-	2.894mA	3mA
2	-	20V	-5.797mA	-6mA
3	10V	20V	-2.894mA	-3mA

• Theoretical calculations:

$$\begin{aligned}
 (i) \quad & \frac{V_1 - 10}{1} + \frac{V_1 - 0}{2.2} + \frac{V_1 - V_2}{1} = 0 \\
 & \frac{V_2 - V_1}{1} + \frac{V_2 - 0}{2.2} + \frac{V_2 - 20}{1} = 0
 \end{aligned}
 \quad \left\{ \begin{array}{l} V_1 = 8.86V \\ V_2 = 11.75V \end{array} \right.$$

$$\therefore \text{Current } (i) = \frac{V_2 - V_1}{1k\Omega} = -2.89mA$$

$$(ii) \quad \left. \begin{aligned} \frac{V_1 - 10}{1} + \frac{V_1 - 0}{2.2} + \frac{V_1 - V_2}{1} &= 0 \\ \frac{V_2 - V_1}{1} + \frac{V_2 - 0}{1} + \frac{V_2 - 0}{2.2} &= 0 \end{aligned} \right\} \begin{aligned} V_1 &= 4.884V \\ V_2 &= 1.99V \end{aligned}$$

$$\therefore \text{current } (i_1) = \frac{V_2 - V_1}{1k\Omega} = +2.89mA$$

$$(iii) \quad \left. \begin{aligned} \frac{V_1 - V_2}{1} + \frac{V_1 - 0}{2.2} + \frac{V_1 - V_2}{1} &= 0 \\ \frac{V_2 - V_1}{1} + \frac{V_2 - 0}{2.2} + \frac{V_2 - 20}{1} &= 0 \end{aligned} \right\} \begin{aligned} V_1 &= 3.98V \\ V_2 &= 9.77V \end{aligned}$$

$$\therefore \text{current } (i_2) = \frac{V_2 - V_1}{1k\Omega} = -5.78mA$$

- By superposition theorem,

$$i_1 + i_2 = +2.89mA - 5.78mA$$

$$\therefore i_1 + i_2 = -2.89mA = i$$

Hence, superposition theorem is verified.

ii) Replacing all other independent current sources with an open circuit (thereby eliminating current i.e., $i=0$; internal impedance of an ideal current source is infinite).

- The theorem is applicable to linear networks (time varying or time invariant) consisting of independent sources, linear dependent sources, linear passive elements, and linear transformers.

- Superposition works on current & voltage, but not power. In other words, sum of powers of each source with other sources turned off is NOT equal to real consumed power.

o Procedure:

- Make connections as per circuit diagram in case (i).
- Set $V_1=10V$ and $V_2=0$ and note down reading of ammeter. (i)
- Now set $V_1=0$ and $V_2=20V$ and note down reading of ammeter. (ii)
- Apply voltages on both sides, and note down ammeter reading. (i)
- The current measured $i = i_1 + i_2$ means that the superposition theorem is verified experimentally.

o Result:

- Superposition Theorem is verified, both theoretically and practically.

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